SWAP VALUES AND PRICES

Learning Outcome

- contrast the value and price of swaps

In the prior lesson, we showed a swap price (or par swap rate) to be a periodic fixed rate that equates the present value of all future expected floating cash flows to the present value of fixed cash flows. The swap rate (s_N for N periods) is equivalent to a forward rate, $F_0(T)$, that satisfies the no-arbitrage condition. Similar to other forwards, the initial contract value (ignoring transaction and counterparty credit costs) is zero ($V_0(T) = 0$).

In contrast to other forward commitments, which involve a *single* settlement at maturity, a swap contract involves a *series* of periodic settlements with a final settlement at contract maturity. Recall from an earlier lesson that the value of a forward contract at maturity from the long forward (or forward buyer's) perspective is $V_T(T) = S_T - F_0(T)$, where S_T is the spot price at maturity and $F_0(T)$ is the forward price. For a swap with periodic exchanges, the current MRR is the "spot" price and the fixed swap rate, s_N, is the forward price. Restating this result for the fixed-rate payer on a swap, the periodic settlement value is

$$\text{Periodic settlement value} = (\text{MRR} - s_N) \times \text{Notional amount} \times \text{Period}. \qquad 4$$

The value of a swap on any settlement date equals the current settlement value in Equation 4 *plus* the present value of all remaining future swap settlements. Although swap market conventions vary, for purposes of this lesson we will assume the MRR sets at the beginning of each interest period, with the same periodicity and day count convention as the swap rate. The fixed versus floating difference is exchanged at the end of each period.

EXAMPLE 3

Esterr Inc. Swap Value and Price

Esterr entered a 3.5-year CAD250 million interest rate swap contract under which it pays a fixed quarterly swap rate of 2.05% and receives three-month CAD MRR. The fixed swap *price* of 2.05% paid by Esterr remains constant over the life of the contract. While the first three-month CAD MRR is *known* at $t = 0$, the remaining 13 MRR settings are *unknown* but are *expected* to equal the respective implied forward rates (IFRs) for each period through maturity. A prior lesson showed how IFRs are derived from zero rates, which were then used to solve for the fixed swap rate by setting the present value of fixed payments equal to the present value of floating payments.

The *value* of Esterr's swap contract will change as time passes and interest rates change. We first consider the passage of time with no change to expected interest rates. That is, the MRR sets each period based on the implied forward rates at trade inception. Assume the following implied forward rates apply to Esterr's swap at $t = 0$:

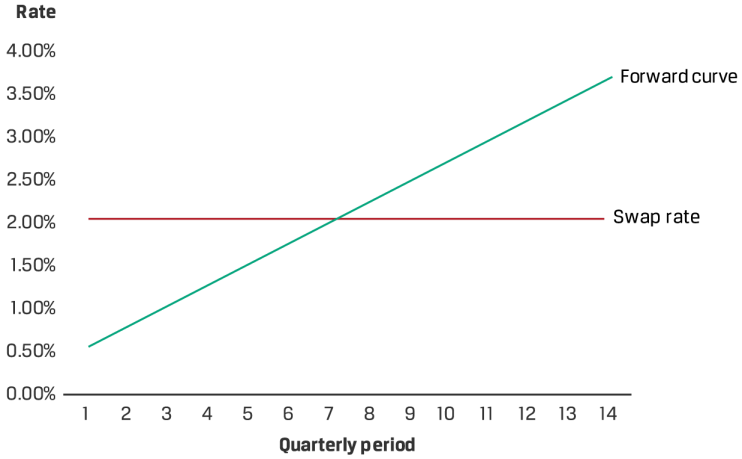
Period	IFR	Rate
1	$\text{IFR}_{0,3m}$	0.50%
2	$\text{IFR}_{3m,3m}$	0.74%
3	$\text{IFR}_{6m,3m}$	0.98%
4	$\text{IFR}_{9m,3m}$	1.22%
5	$\text{IFR}_{12m,3m}$	1.46%

7	$IFR_{18m,3m}$	1.94%
8	$IFR_{21m,3m}$	2.18%
9	$IFR_{24m,3m}$	2.43%
10	$IFR_{27m,3m}$	2.67%
11	$IFR_{30m,3m}$	2.91%
12	$IFR_{33m,3m}$	3.15%
13	$IFR_{36m,3m}$	3.39%
14	$IFR_{39m,3m}$	3.78%

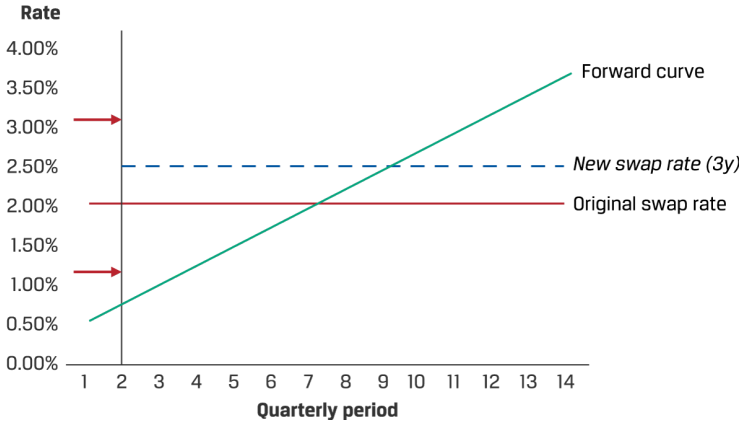
Based on these forward rates, Esterr expects to *make* a net swap payment each quarter through the seventh period—since (MRR – s_N) = (1.94% – 2.05%) < 0—and *receive* a net quarterly swap payment starting in the eighth period, where (MRR – s_N) = (2.18% – 2.05%) > 0. Using Equation 1, consider the periodic settlement values for the first two periods from Esterr's perspective as the fixed-rate payer:

- **Period 1:** –CAD968,750 = (0.5% – 2.05%) × CAD250 million × 0.25.
- **Period 2:** –CAD818,750 = (0.74% – 2.05%) × CAD250 million × 0.25.

What is the swap MTM *value* from Esterr's perspective immediately after the second periodic settlement if forward rates remain unchanged? Note that the swap *price* (the fixed swap rate of 2.05%) was set at inception to equate the present value of fixed versus floating payments.



As we move forward in time with *no* change to interest rate expectations, the present value of remaining floating payments rises above the present value of fixed payments at 2.05%, as Esterr has made a net settlement payment in the first two periods.



As the diagram shows, as time progresses, Esterr realizes an MTM gain on the swap, since

$$\Sigma \text{ PV(Floating payments received)} > \Sigma \text{ PV(Fixed payments paid)}.$$

If we instead consider interest rate changes only, from Esterr's perspective as the fixed-rate payer (and floating-rate receiver), we can show the conditions under which Esterr has a swap MTM gain or loss:

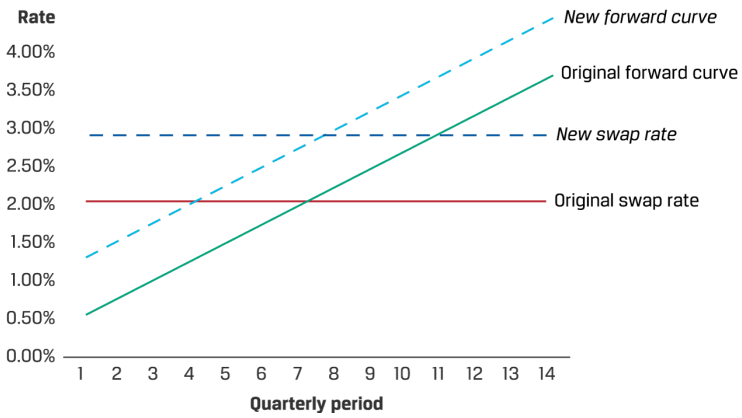
- Esterr realizes an MTM *gain* on the swap as the fixed-rate payer if

  Σ PV(Floating payments received) > Σ PV(Fixed payments paid).

- Esterr realizes an MTM *loss* on the swap as the fixed-rate payer if

  Σ PV(Floating payments received) < Σ PV(Fixed payments paid).

A rise in the expected forward rates after inception will increase the present value of floating payments, while the fixed swap rate will remain the same. We show the effect of an immediate change in interest rates following trade inception in the following diagram:



The new forward curve in this diagram is composed of higher IFRs. If we were to now solve for a fixed swap rate using these higher IFRs by setting floating and fixed cash flows equal to one another, the new swap rate would be above the original swap rate. However, since Esterr has locked in future fixed payments at the *lower* original swap rate while receiving higher expected future MRRs, the swap has a positive MTM value to Esterr.

Another interpretation of an interest rate swap is that the fixed-rate payer (floating-rate receiver) is *long* a floating-rate note (FRN) priced at the MRR and *short* a fixed-rate bond with a coupon equal to the fixed swap rate. Note that the combination of long and short bond positions causes both the purchase and sale prices of the two bonds at inception and the return of principal at maturity cancel one another out, and only the fixed versus variable coupon payments remain. The following example shows how the change in an interest rate swap's value is similar to that of a fixed-income security using an earlier investor swap example.

EXAMPLE 4

Fyleton Investments

Fyleton Investments entered a five-year, receive-fixed GBP200 million interest rate swap in an example from an earlier lesson to increase the duration of its fixed-income portfolio. Assume in this case that Fyleton receives a fixed swap rate of 2.38% on a semiannual basis versus six-month GBP MRR. Further assume that initial six-month GBP MRR sets at 0.71% and, as in the case of Esterr, the MRR forward curve is upward sloping. How will the value of Fyleton's swap change as time passes and interest rates change?

is

GBP1,670,000 = (2.38% – 0.71%) × GBP200 million × 0.5.

What is the MTM value from Fyleton's perspective immediately following the first settlement if implied forward rates remain the same as at trade inception?

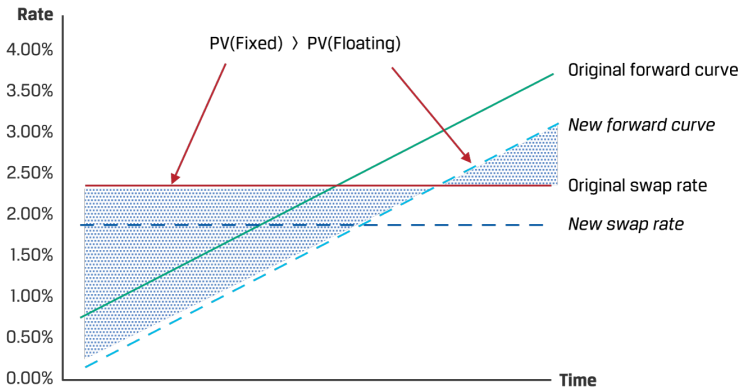
As the fixed-rate *receiver*, Fyleton will realize an MTM *loss* on the swap, since

Σ PV(Floating payments paid) > Σ PV(Fixed payments received).

Second, consider a change in forward rates. A *decline* in expected forward rates immediately following trade inception will *reduce* the present value of floating payments, while the fixed swap rate will remain the same. Fyleton will realize an MTM gain as the fixed-rate receiver, since

Σ PV(Fixed payments received) > Σ PV(Floating payments paid).

This MTM gain is shown by the different size of the shaded areas under the original swap rate using the new forward curve in the following diagram.



Note the similarity between the receive-fixed swap exposure profile and that of a long cash fixed-rate bond position. In the first instance, we would expect a fixed-rate par bond to be priced at a *discount* as time passes if rates rise as per an upward-sloping forward curve, while an FRN priced at the MRR would remain at par. In the second instance, a decline in implied forward rates (or downward shift in the forward curve) would cause a fixed-rate par bond to be priced at a *premium*, while the FRN price would not change. We will explore how term structure and yield curve changes affect swap values and bond prices in greater detail later in the curriculum.

Question Set: Swap Prices and Values

1. Determine the correct answers to fill in the blanks: A rise in the expected forward rates after inception will _____ the present value of floating payments, causing a fixed-rate receiver to realize a(n) _____ in MTM value on the swap contract.

   Hide Solution

   Solution:

   A rise in the expected forward rates after inception will *increase* the present value of floating payments, causing a fixed-rate receiver to realize a *decline* in MTM value on the swap contract.

2. Identify the following statement as true or false, and justify your response: The fixed rate on an interest rate swap is priced such that the present value of the floating payments (based on respective implied forward rates for each period) is equal to the present value of the fixed payments at $t = 0$.

This statement is true. The fixed rate on an interest rate swap may be solved by setting the present value of floating payments equal to the present value of fixed payments. We use zero rates to derive implied forward rates, or breakeven reinvestment rates, between a shorter and a longer zero rate. These IFRs represent the respective floating rates that are expected to apply for each future period at time $t = 0$.

3. Identify which of the following statements is associated with which position in an interest rate swap contract.

1. Establishes a set of certain net future cash flows on a swap contract at inception	A. Fixed-rate payer (Floating-rate receiver)
2. Realizes an MTM gain on a swap contract if the expected future floating-rate payments increase	B. Fixed-rate receiver (Floating-rate payer)
3. An investor may increase portfolio duration by entering this position in a swap contract	C. Neither a fixed-rate payer nor a fixed-rate receiver

Hide Solution

Solution:

1. C is correct. A swap contract establishes a set of *certain* fixed future cash flows that are exchanged for a set of *expected* (*uncertain*) floating future cash flows. Therefore, *neither* a fixed-rate payer nor a fixed-rate receiver knows the *net* future cash flows of a swap at inception.
2. A is correct. A fixed-rate payer realizes an MTM gain on a swap contract if the expected future floating-rate payments increase.
3. B is correct. A fixed-rate receiver may increase portfolio duration by entering this position on a swap, because receiving fixed is similar to a long bond position.

4. Determine the correct answers to fill in the blanks: The value of a swap on any settlement date equals the _____ settlement value plus the present value of all remaining _____ swap settlements.

Hide Solution

Solution:

The value of a swap on any settlement date equals the *current* settlement value plus the present value of all remaining *future* swap settlements.

5. Describe how an investor may use an interest rate swap to reduce the duration of a fixed-income portfolio, and justify your response.

Hide Solution

Solution:

A pay-fixed swap is similar to a short bond position that reduces duration, because the fixed-rate payer (floating-rate receiver) is effectively long a floating-rate note priced at the MRR and short a fixed-rate bond with a coupon equal to the fixed swap rate.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

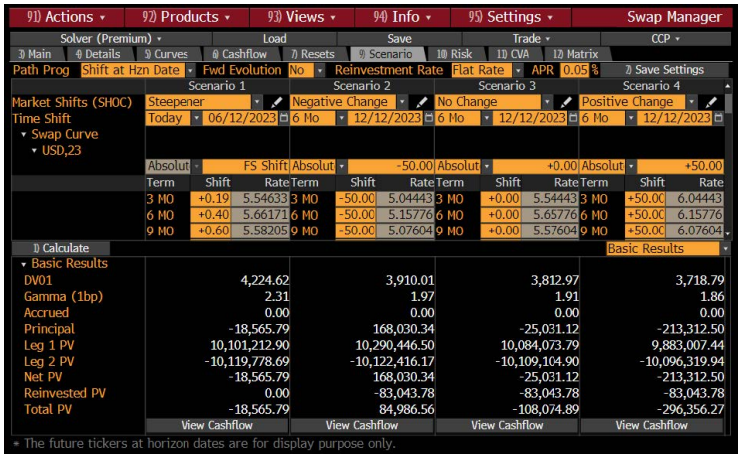
Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

The Bloomberg SWPM function allows you to view how a plain-vanilla interest rate swap is priced.

> "SWPM <GO> allows you to price over-the-counter derivative contracts, so you can accurately quantify your market exposures for a wide range of vanilla and exotic interest-rate swaps, interest-rate options, swaptions, and interest-rate and hybrid structured notes. Using an exhaustive list of pre-trade analytical tools, you can analyze and update curves and cash flows, as well as perform risk and scenario analyses for the entire deal or individual legs." -Bloomberg

To explore this function, type SWPM into the Bloomberg command line and press Enter or <GO>. Click on the gray "Scenario" tab. Click on the "Market Shifts (SHOC)" dropdown menu. Then choose a scenario, such as steepener, flattener, down shifts, and up shifts, which represent changes in the yield curve. Once you've chosen a scenario, click on the gray "Calculate" button in the middle left of the screen to see the present value of swaps to address various curve shifts.



© 2023 Bloomberg L.P.

Please provide feedback on this Bloomberg feature.

Discuss Filter

Discussion

What do you want to discuss... (required)

MP I think there is an errata in the example of Fyleton, it first says at inception Fyleton earns profit(makes total sense, since he receives a higher rate than what he is supposed to pay) And a parragraph after, Fyleton it is seen as if it was incurring an MTM loss (which makes no sense due to the previous mentioned fact)

Created 4 months ago by Manuel Paredes Caldito 1 reply | Last Activity: 2 months ago

Show All replies Reply to this Comment

AZ Σ PV(Floating payments paid) > Σ PV(Fixed payments received).
--> please fix it to (payments "to be received") as its confusing like that

Created 3 months ago by Abdulaziz Zahid 0 replies | Last Activity: 3 months ago

Reply to this Comment

© 2025 CFA Institute. All Rights Reserved.

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search

Home
Study Plan
Lessons
Flashcards
Practice
Mock Exams
Game Center
Discussions
Search